Atomwise

Bringing the Power of AI Drug Discovery to our Strategic Partners



www.Atomwise.com

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Better Medicines, Faster.

The leader in AI for drug discovery

1st
to invent and use ConvNets for drug design

$50M+
funding raised from prominent investors

16B+
small molecules in AtomNet

700+
drug discovery projects to date

Top10
we work with the world's top pharma companies

75%
success across AIMS projects to date

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Partnerships that Accelerate Drug Discovery

Over $5.5B in signed deals

















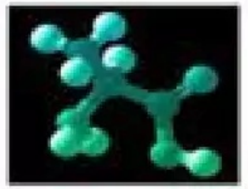
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Leading AI Technology for Drug Design

Screen 16B+ small molecules with AtomNet®

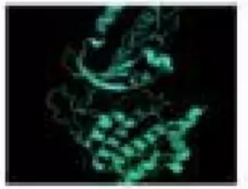
The 1st company to use Convolutional Neural Nets (CNNs) for structure-based drug discovery

1



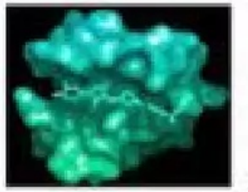
Map Atoms in 3D Space, use Voxels and Poses

2

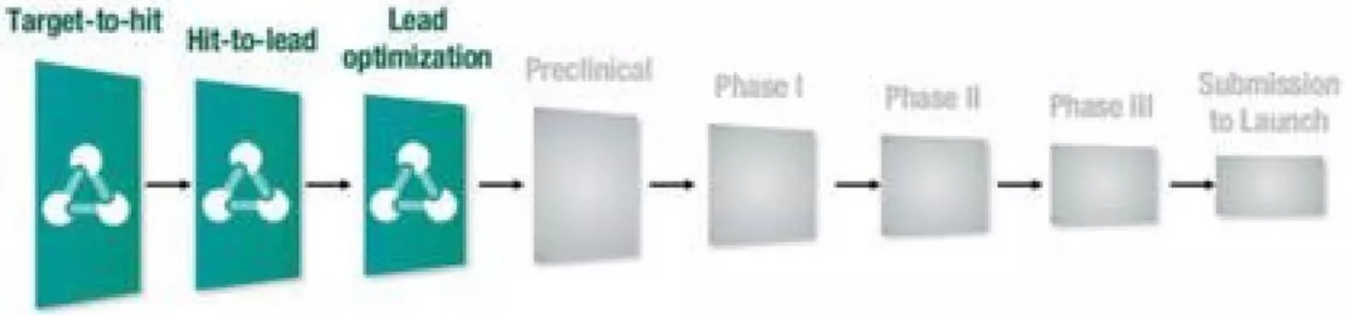


Screen 16+B chemicals for predicted interaction with the target

3



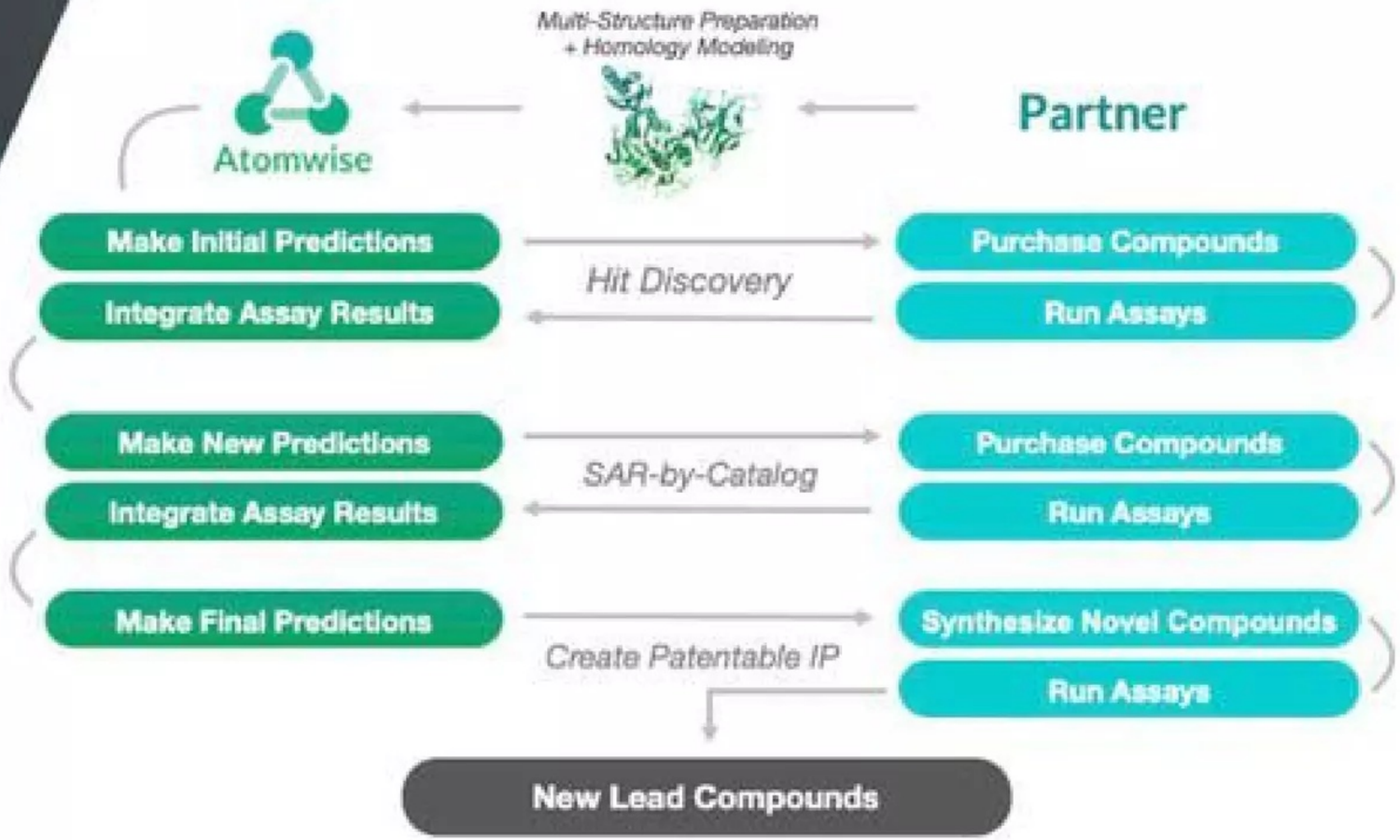
Send compounds to partner for lab testing (75% success)



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A Seamless Partnership from Start to Finish

With our team of Med Chem and AI experts



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Proven Success with Challenging Targets

We solve the hardest problems in computer aided drug design, delivering hits even when there is little-to-no structural data for a target. We have a history of success, delivering at least one hit validated in laboratory experiments 74% of the time, keeping our partners ahead of their competition.



Overall Success

120 Projects

74% Success



No Training Data

55 Projects

71% Success



Homology Model

20 Projects

71% Success



X-ray Structure

84 Projects

75% Success



Protein-Protein Interactions

23 Projects

83% Success



EM-Cryo Data

1 Project

100% Success

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Academic Partnerships

Several ways to collaborate with Atomwise

AIMS Awards Program



Fast and streamlined project to conduct hit discovery



Flexible with publication and commercialization

Co-Development Project



Tailored collaboration spanning from hit discovery to identification of a clinical candidate



Revenue sharing model that reflects each party's contribution

Joint Venture



Sharing in the risk and benefits of developing a drug candidate



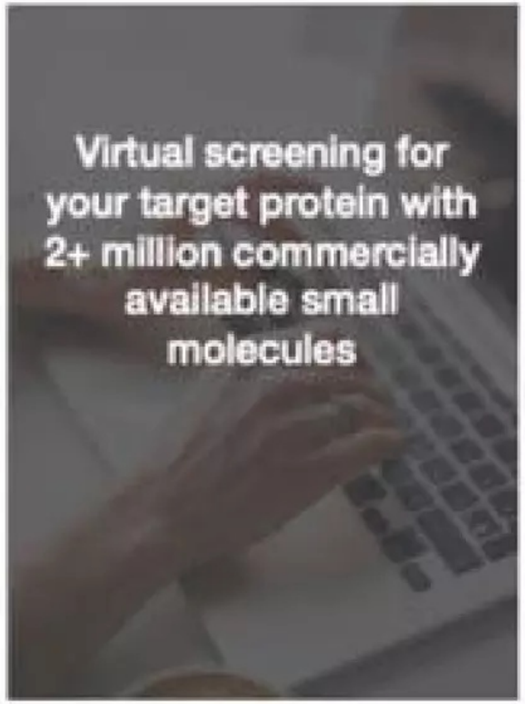
Access to our CADDS ecosystem, a network of Atomwise partnerships that can support our JVs

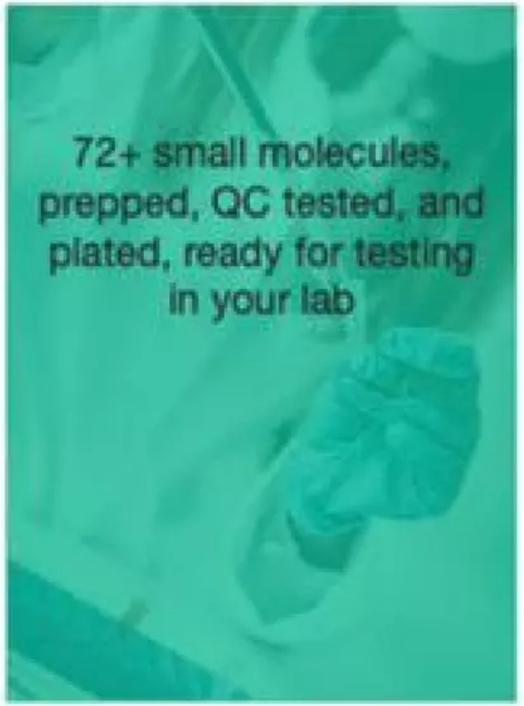
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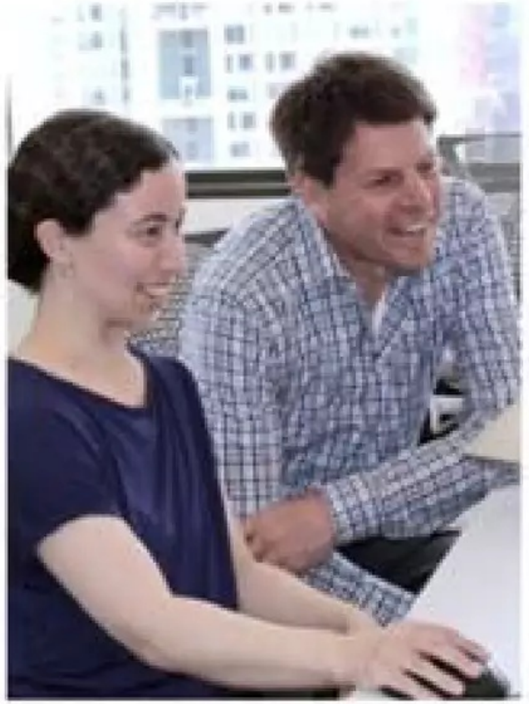
AIMS Awards Program

Artificial Intelligence Molecular Screen (AIMS) Awards provide valuable access to AI-based drug design to academic research labs.





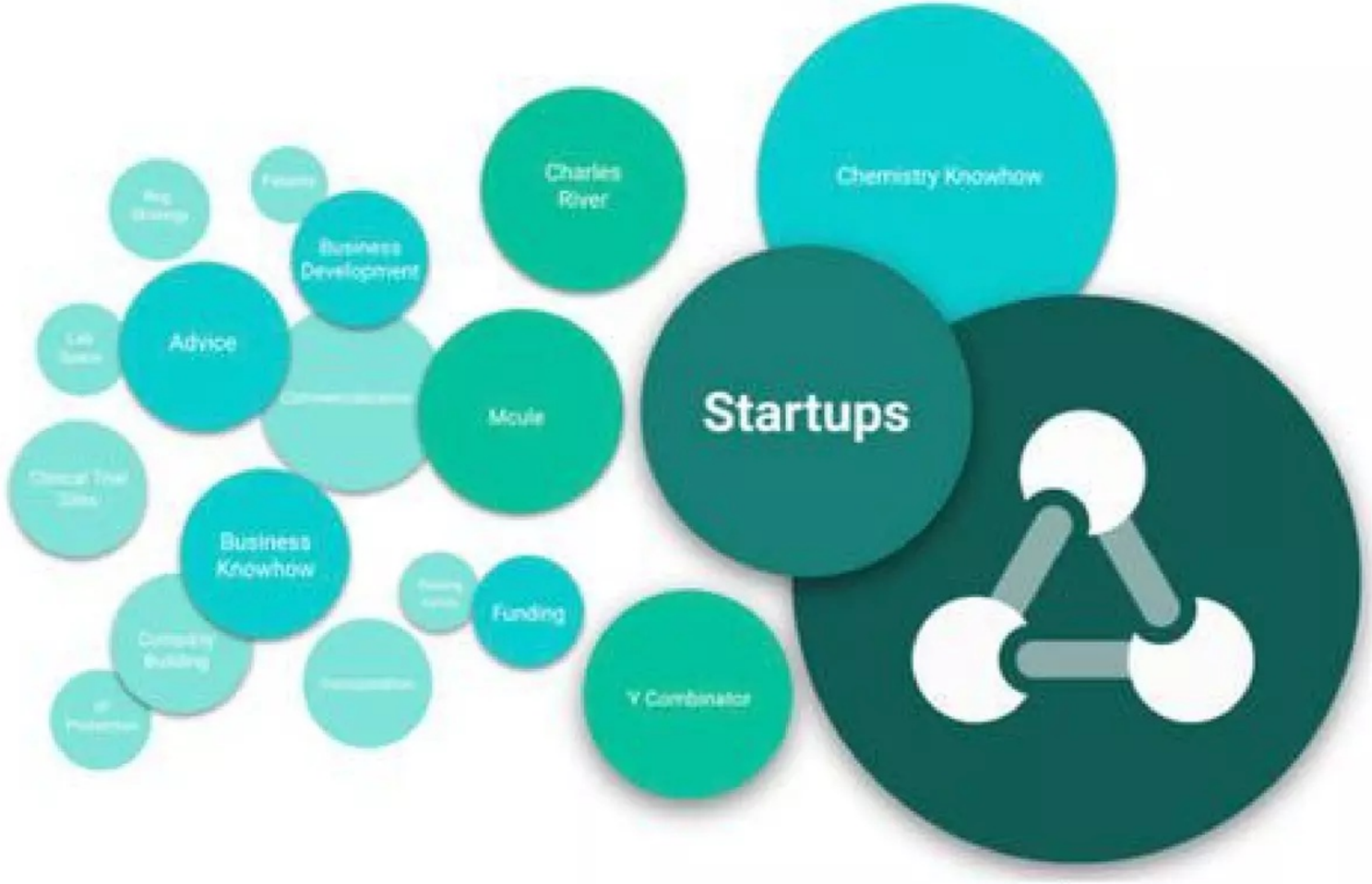




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Our CADDS Ecosystem

Chemistry for Academic Drug Discovery Startups Ecosystem



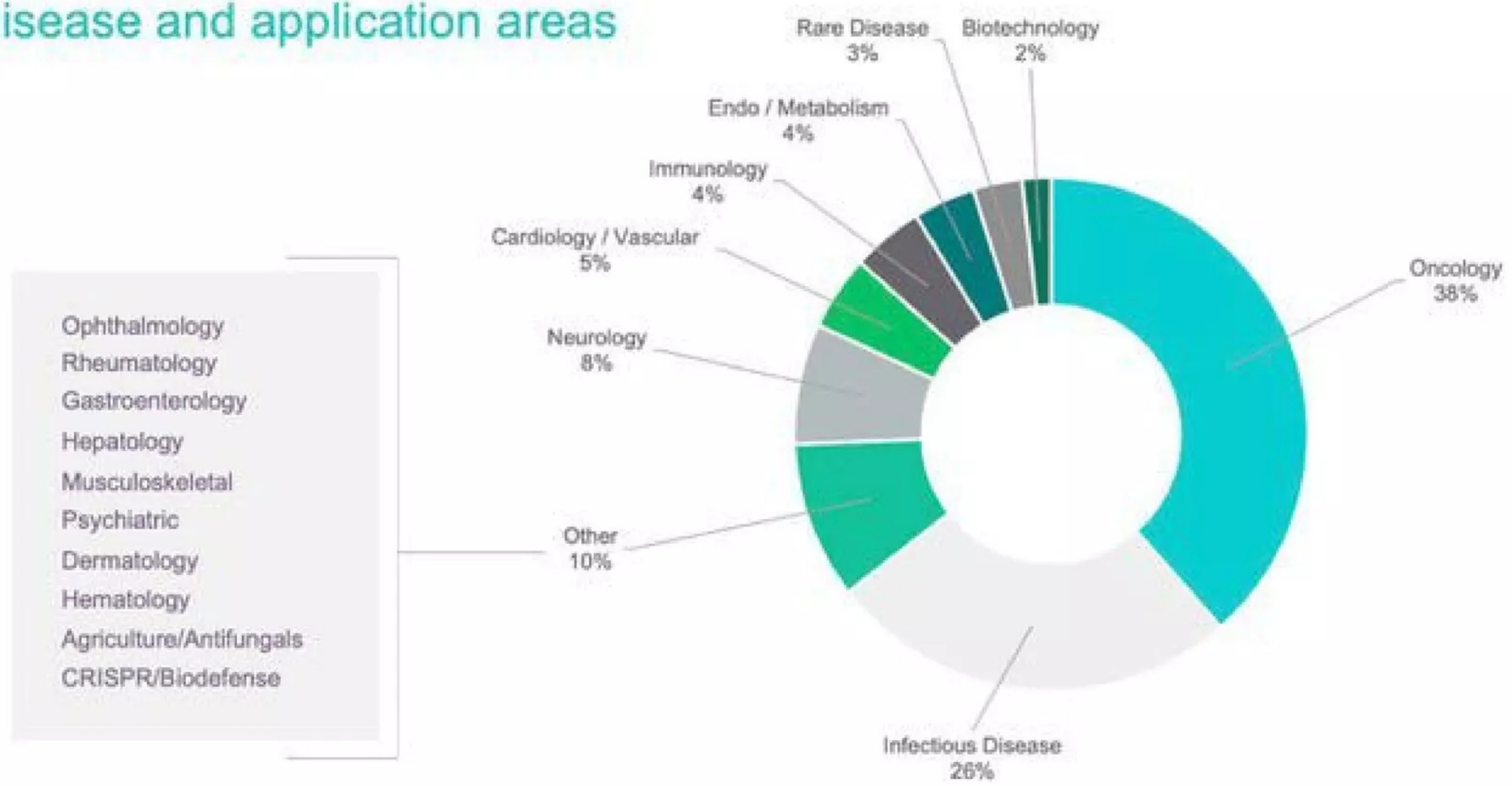
Supporting the full lifecycle of preclinical drug development

Startups working with Atomwise have access to all the tools and resources that Atomwise and our partners, including leading CRO, Charles River, and vendors, Mcule and Enamine, have to offer to support innovation and the development of the next generation of medicines.

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Enormous Breadth of Applications

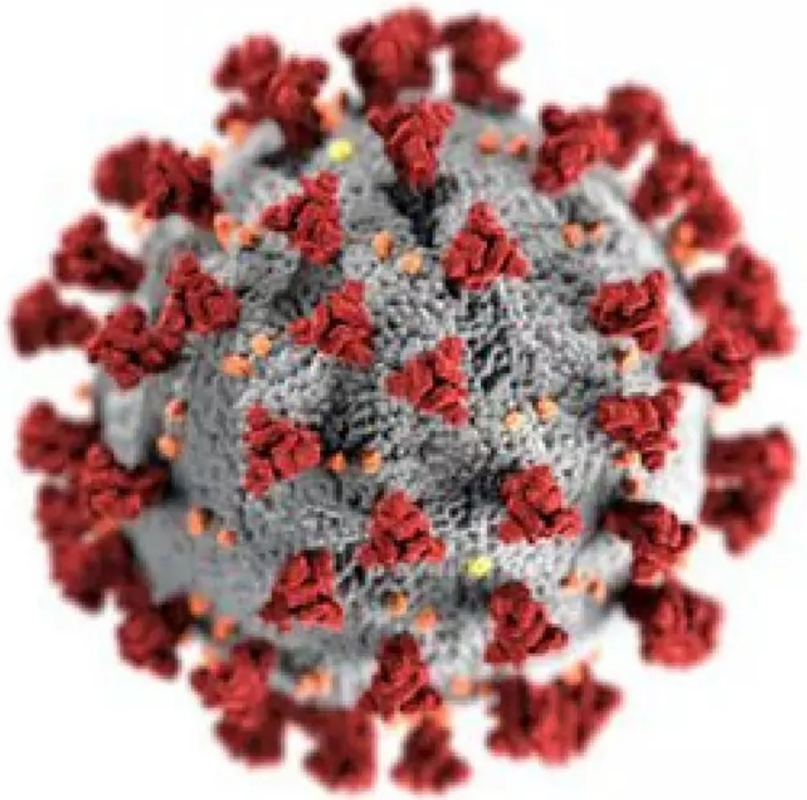
775 accepted projects to date across diverse disease and application areas



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Broad Spectrum COVID Projects

Targeting MERS, SARS-CoV-1 and SARS-CoV-2 with multiple angles of attack



Target	Institute
Undisclosed	Dana-Farber Cancer Institute
Nucleocapsid (N-protein)	Columbia University
NSP15	University of Connecticut
Papain-Like Protease (PLpro)	University of Kentucky
IL-6 Signaling Pathway	University of Manitoba
RdRp in NSP12	University of Manitoba
Spike-ACE2	University of South Australia
Spike-ACE2	University of Texas Medical Branch (UTMB) Health
Spike (heptad region)	Jazan University, Saudi Arabia
Undisclosed	University of Texas Health Science Center San Antonio
Undisclosed	University of Toledo

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From Gene Variant to Potential Cure

HTRA1 - 1st in class treatment for Age-Related Macular Degeneration (AMD)

"Theia Biosciences is building off of the initial research started in collaboration with Atomwise during my first AIMS Award. Theia Biosciences is focused initially on meeting an unmet medical need in helping treat age-related macular degeneration through developing small molecule inhibitors."



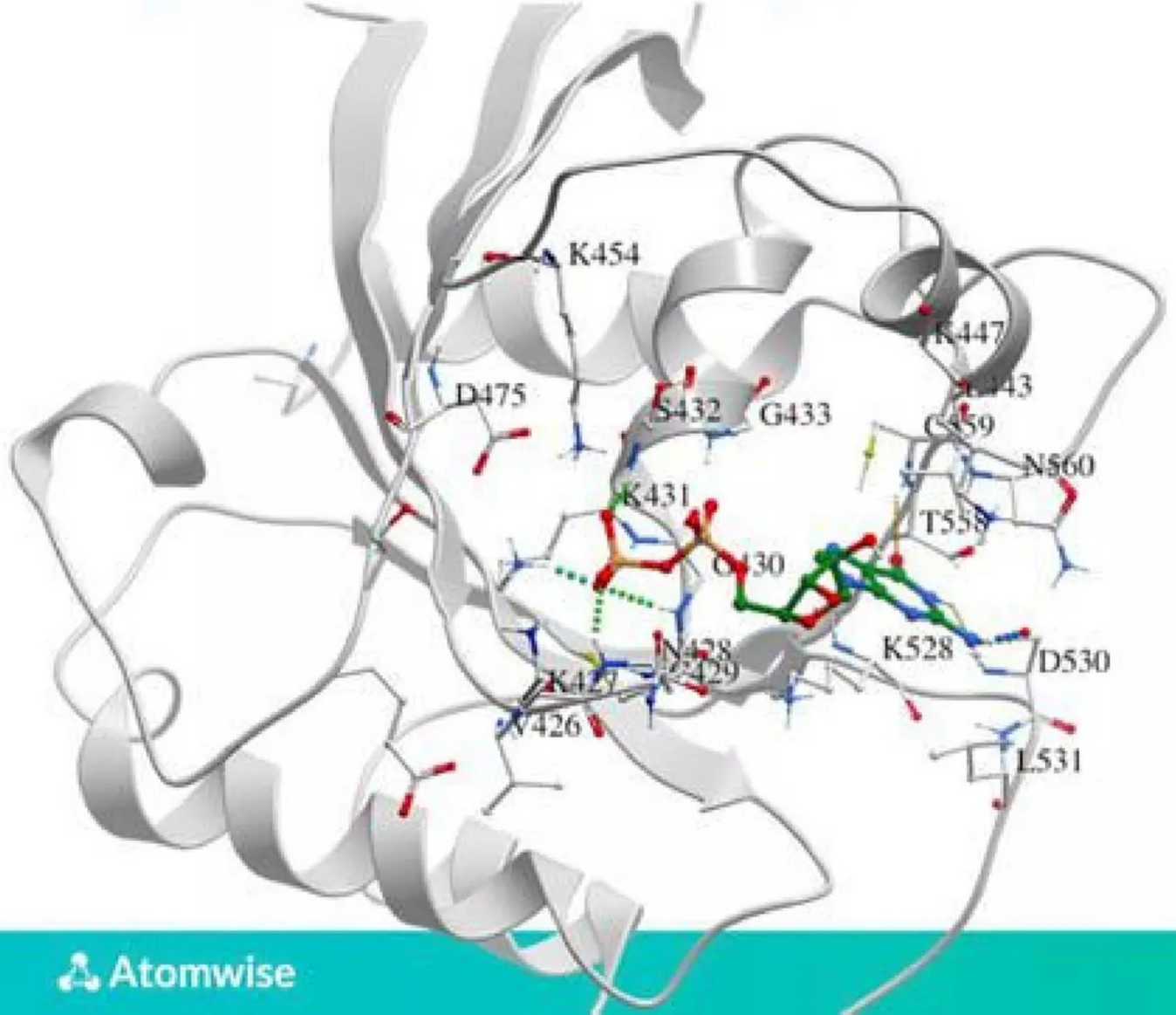
Dr. Jeff Perry
Assistant Professor of Biochemistry at UC Riverside and Scientific Co-founder of Theia Biosciences



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Parkinson's Disease

Targeting Miro1 and Neurodegeneration



Short Article

Cell Metabolism

Miro1 Marks Parkinson's Disease Subset and Miro1 Reducer Rescues Neuron Loss in Parkinson's Models

Chung-Han Hsieh,[1,2] Li Li,[1,2] Roeland Vanhauwaert,[1] Kong T. Nguyen,[1] Mary D. Davis,[1] Guojun Bu,[3] Zbigniew K. Wszolek,[4] and Xinnan Wang[1,2,5]
[1]Department of Neurosurgery, Stanford University School of Medicine, Stanford, CA 94305, USA
[2]Atomwise Inc., San Francisco, CA 94105, USA
[3]Department of Neuroscience, Mayo Clinic, Jacksonville, FL 32224, USA
[4]Department of Neurology, Mayo Clinic, Jacksonville, FL 32224, USA
[5]These authors contributed equally
[6]Lead Contact
[7]Correspondence: xinnanw@stanford.edu
https://doi.org/10.1016/j.cmet.2019.08.023

Highlights

- Atomwise found a novel small molecule that promotes Miro1 degradation in PD fibroblasts
- Treating PD models with this compound rescues dopaminergic neurodegeneration
- Miro1 marker and engaging in Miro1-based therapies could open new avenues to personalized medicine

Hsieh et al., 2019, Cell Metabolism 30, 1–15
December 3, 2019 © 2019 Elsevier Inc.
https://doi.org/10.1016/j.cmet.2019.08.023

CellPress

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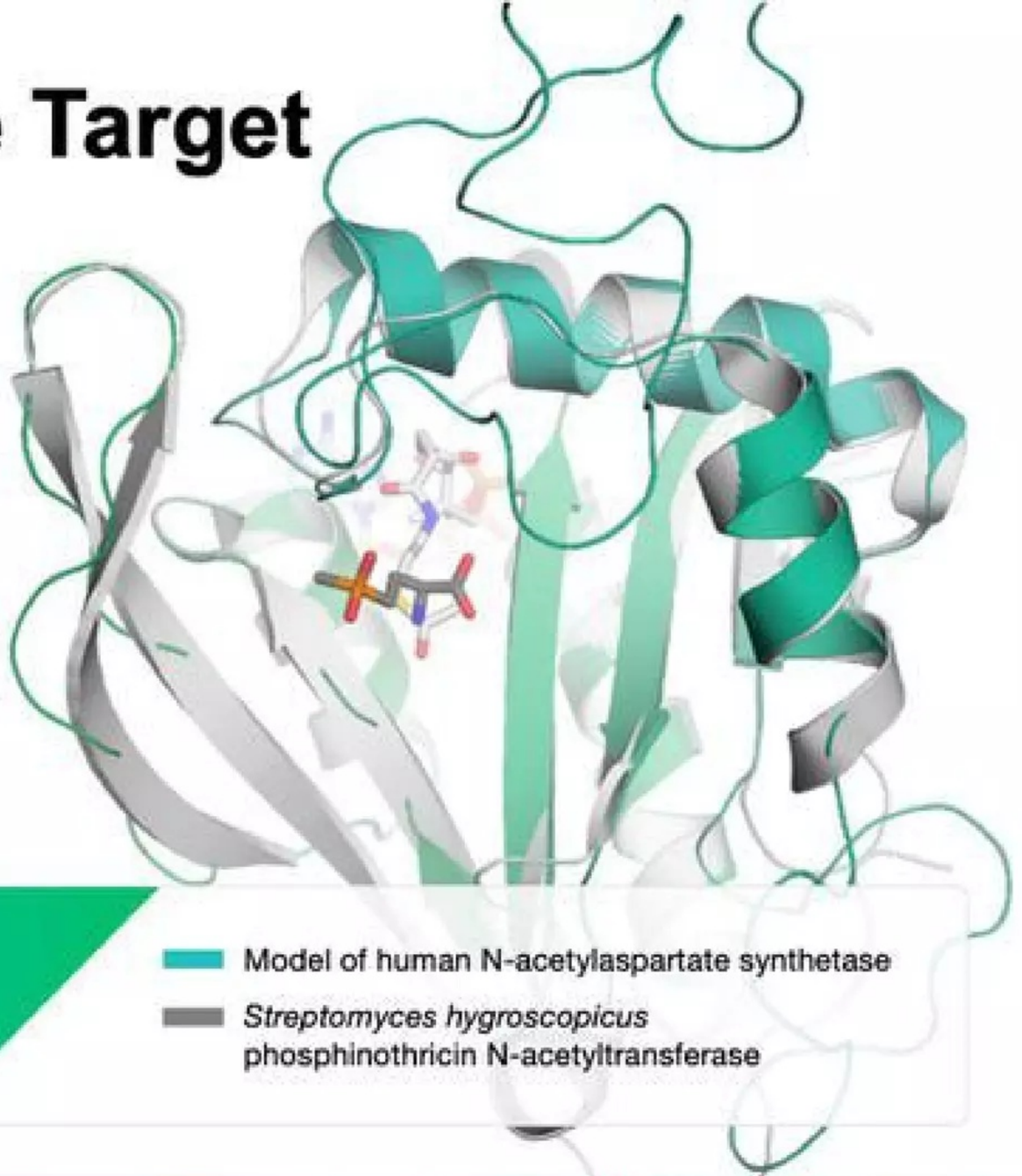
Hope for an Undruggable Target

N-acetylaspartate synthetase (NAT8L)

No Physical screening
- Membrane-associated protein and difficult to purify

No Medicinal Chemistry
- No drug-like inhibitors were previously known

No Computational screening
- No available crystal structures
- Extremely low sequence identity template (20%)



Model of human N-acetylaspartate synthetase

Streptomyces hygroscopicus phosphinothricin N-acetyltransferase

Atomwise screened 7.2M compounds in <2 hrs
5/60 compounds identified as hits.

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